

MARAVEH TAPPEH, Iran, Islamic Republic of

WMO: 407210

Lat: 37.8009N

Lon: 55.9434E

Elev: 460

StdP: 95.92

Time Zone: 3.50 (IRN)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.3	-0.5	-11.6	1.5	5.0	-8.4	2.0	8.5	19.6	11.9	16.1	9.5	3.1	260	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	39.4	20.6	37.6	20.5	36.1	20.6	25.0	33.5	23.9	31.8	23.0	31.1	4.0	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.4	18.1	30.8	21.4	17.0	29.2	20.5	16.0	28.1	79.4	33.6	74.5	31.6	70.7	31.0	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 13.5	11.0	9.5	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-6.2	25.4	3.1	1.7	-8.4	26.6	-10.2	27.6	-12.0	28.6	-14.2	29.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	8.1	7.5	12.5	16.4	23.1	27.7	29.7	29.5	25.6	19.1	12.5	9.5
	DBStd	9.10	4.03	4.55	5.19	4.72	3.73	3.26	2.89	2.92	2.91	4.67	5.16	4.31
	HDD10.0	312	88	93	31	4	0	0	0	0	1	31	63	
	HDD18.3	1418	318	304	193	92	4	0	0	1	48	182	275	
	CDD10.0	3414	28	22	108	196	406	532	612	605	469	282	106	47
	CDD18.3	1478	0	1	12	33	152	282	353	347	220	70	7	1
	CDH23.3	16903	0	1	124	293	1459	3394	4613	4467	2013	503	36	1
	CDH26.7	8636	0	0	38	99	649	1824	2588	2447	808	176	7	0
(14) Wind	WSAvg	3.9	4.7	4.1	4.0	3.6	3.9	4.2	4.1	4.0	3.7	3.4	3.3	4.3
(15) Precipitation	PrecAvg	412	39	45	58	53	33	15	20	21	17	31	40	37
	PrecMax	629	86	79	121	115	114	55	92	63	75	92	81	69
	PrecMin	223	12	16	21	7	2	0	2	0	0	4	6	6
	PrecStd	95	18	19	22	29	25	14	21	16	18	18	19	17
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.9	22.0	31.1	33.3	37.8	40.7	42.2	40.9	37.0	35.0	27.2	22.9
		MCWB	11.0	11.6	15.4	17.6	19.8	20.1	22.0	20.7	18.8	16.9	15.1	11.1
	2%	DB	18.2	18.6	26.5	29.4	34.7	38.1	39.7	38.7	34.0	30.8	23.4	20.4
		MCWB	9.1	9.9	14.0	17.2	18.3	20.2	21.2	20.3	19.2	16.5	12.3	10.1
	5%	DB	16.3	16.3	23.6	26.6	32.7	36.2	37.6	36.8	32.5	28.1	20.9	18.0
		MCWB	8.1	8.6	12.8	16.0	18.1	20.3	21.3	20.0	19.0	16.0	11.6	9.1
	10%	DB	14.3	14.2	20.7	24.1	30.4	34.5	35.8	35.3	31.0	26.0	19.0	16.2
		MCWB	7.2	8.0	12.0	14.9	18.1	19.8	21.9	20.2	18.8	15.7	11.4	8.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	12.4	16.8	19.9	21.4	24.0	26.4	25.6	23.0	19.8	15.8	12.3
		MCDB	18.8	20.0	27.4	30.8	30.5	32.8	36.5	34.3	31.3	27.3	23.3	19.9
	2%	WB	9.8	10.6	15.0	18.2	20.5	22.9	25.6	24.5	22.0	18.7	14.6	11.0
		MCDB	16.1	16.5	24.3	26.8	29.8	33.3	34.9	32.2	29.3	25.4	21.4	17.3
	5%	WB	8.9	9.5	13.6	17.0	19.6	22.0	24.6	23.5	21.3	17.8	13.5	10.2
		MCDB	14.4	14.4	21.7	24.7	29.3	32.7	33.0	30.9	28.8	24.9	18.4	16.0
	10%	WB	7.9	8.6	12.4	15.8	18.8	21.1	23.6	22.6	20.5	16.9	12.3	9.3
		MCDB	13.3	12.8	19.7	22.6	27.9	31.5	31.5	29.9	28.1	23.7	16.6	14.3
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	7.2	8.7	9.0	10.2	10.8	10.1	9.6	8.9	8.3	7.1	6.6
		MCDBR	9.1	10.4	13.0	13.3	13.7	13.3	12.6	11.7	10.9	11.3	9.2	8.9
	5% WB	MCWBR	4.4	5.7	5.7	6.5	4.9	4.5	4.7	4.3	4.6	4.7	4.3	4.5
		MCWBR	8.6	9.1	11.3	12.2	11.8	11.7	10.4	9.4	8.9	8.9	7.6	8.2
(40) Clear-Sky Solar Irradiance	taub		0.356	0.400	0.467	0.524	0.528	0.524	0.546	0.491	0.464	0.456	0.392	0.357
		taud	2.223	2.089	1.923	1.838	1.867	1.892	1.852	1.975	2.013	2.003	2.151	2.226
	Ebn at Noon		802	808	786	766	772	774	752	785	780	734	747	769
		Edh at Noon	104	135	176	204	201	196	203	175	159	146	111	97
(44) All-Sky Solar Radiation	RadAvg		2.33	3.06	4.13	5.18	6.44	7.53	7.25	6.82	5.60	4.00	2.75	2.12
	RadStd		0.16	0.29	0.29	0.58	0.34	0.28	0.34	0.42	0.20	0.20	0.16	0.18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	N/A	+0.58	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page